

Renewable energy

RENEWABLE ENERGY SOURCES ARE AN ACTIVE AREA OF RESEARCH WORLDWIDE.

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The Sun

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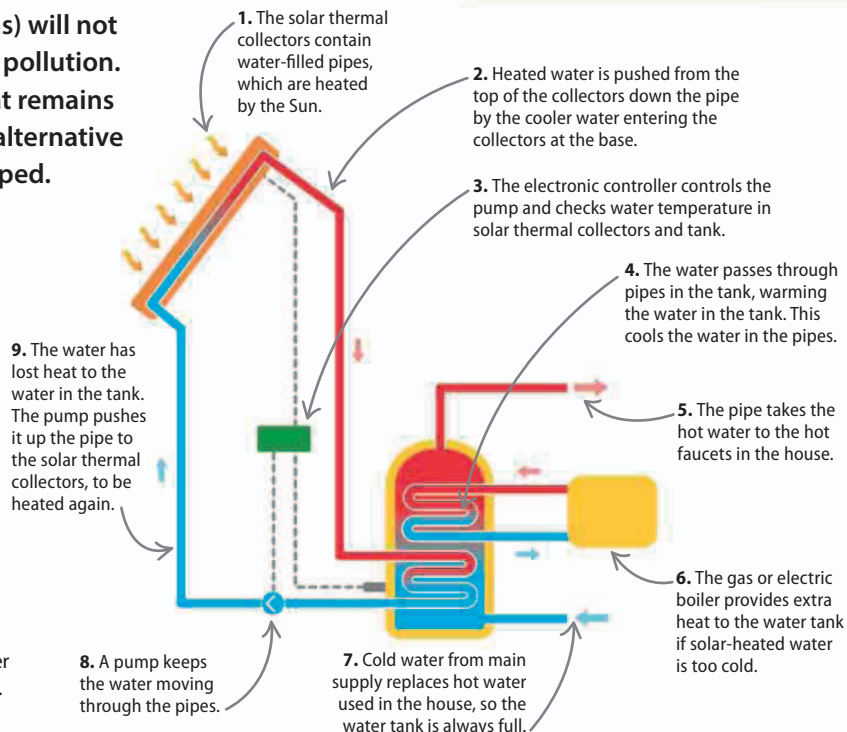
Fossil fuels (coal, oil, and natural gas) will not last forever, and they cause serious pollution. Nuclear energy produces waste that remains dangerous for many centuries. So, alternative sources of energy are being developed.

Solar energy

There are two main ways to convert sunlight into usable energy. The first way is in a solar thermal collector, where a liquid is heated by being pumped through sunlit pipes, and then used to heat a boiler. The second way is to turn it into electrical power by means of photovoltaic cells. These contain material such as the element selenium, which produces an electrical voltage when light falls on it.

► Heat from the Sun

Solar thermal collectors can be used to heat water for warming a house, as shown in this illustration.



Wind turbines

A wind turbine converts the motion of the wind into electrical power. Many wind turbines are grouped into arrays and these are often offshore, where conditions are windier. As winds do not always blow, the amount of energy from wind farms is variable. Wind farms can be unpopular because of their appearance and the noise they make, so careful siting is essential.

► Inside a wind turbine

When the wind turbine blades rotate, a generator (dynamo) produces electricity. A system of gears converts the relatively slow spin of the blades into a more rapid rotation in the dynamo, producing more electrical power.

